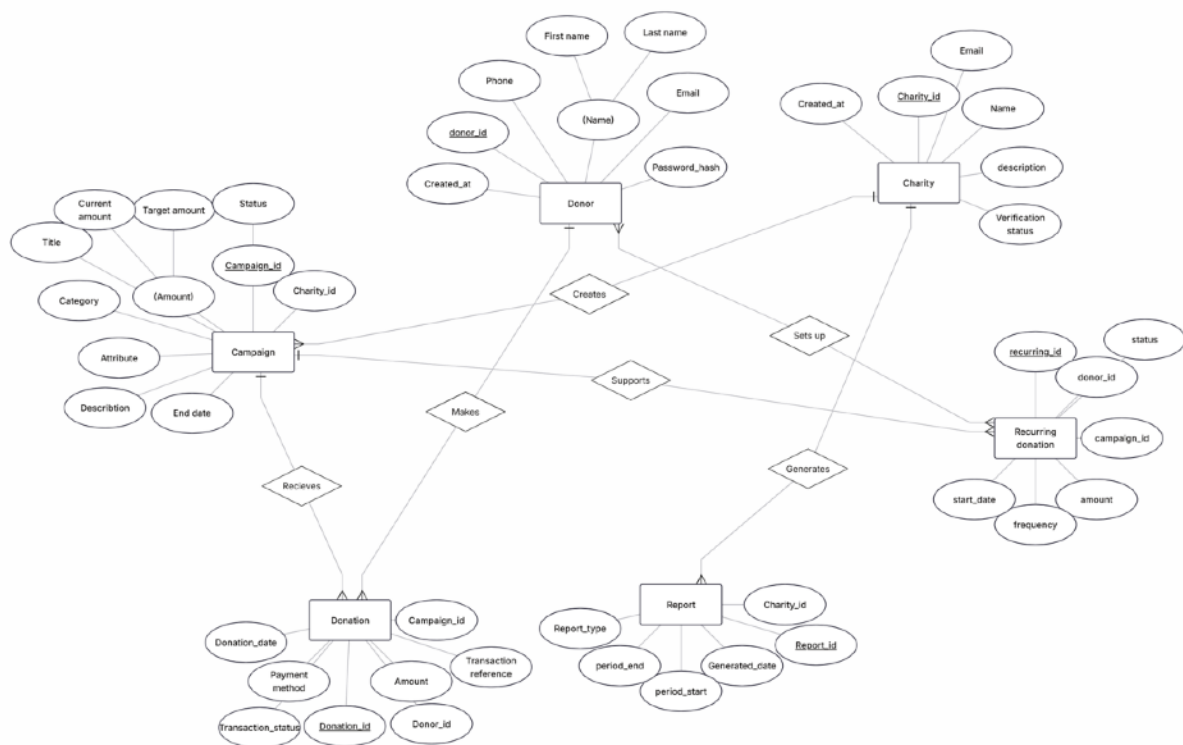


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24 December 2025

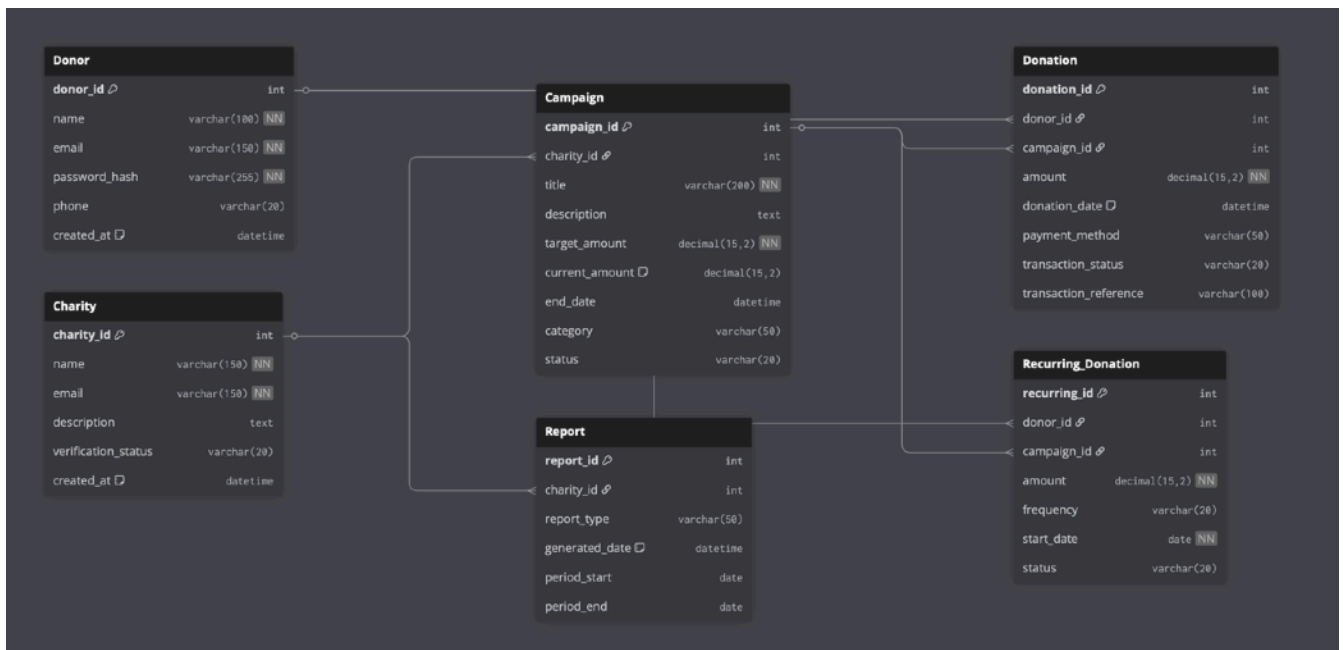
Milestone 2

CCAS 4.3 Donation System

a. Entity-Relationship Diagram (ERD):



b. Relational Database Schema



i. Insert Data (Adding New Records)

QUR_001: Register a New Donor

Purpose: Adds a new donor to the system when they sign up.

Context: Used during the registration process

```
INSERT INTO Donor (name, email, password_hash, phone)
```

```
VALUES ('John Doe', 'john.doe@example.com', 'hashed_password_123', '+201234567890');
```

QUR_002: Create a New Charity Campaign

Purpose: Allows a charity to publish a new fundraising campaign.

Context: Used by Charity Administrators

```
INSERT INTO Campaign (charity_id, title, description, target_amount, end_date,  
category, status)
```

```
VALUES (1, 'Clean Water for All', 'Providing clean water wells to remote villages.',  
50000.00, '2025-12-31 23:59:59', 'Health', 'Active');
```

QUR_003: Process a New Donation

Purpose: Records a successful donation transaction.

Context: Executed after payment confirmation.

```
INSERT INTO Donation (donor_id, campaign_id, amount, payment_method,  
transaction_status, transaction_reference)
```

```
VALUES (1, 1, 150.00, 'Credit Card', 'Completed', 'TXN_987654321');
```

ii. Update Data (Modifying Records)

QUR_004: Update Campaign Progress (After Donation)

Purpose: Updates the current_amount of a campaign whenever a new donation is received.

Context: Keeps campaign progress bars accurate.

```
UPDATE Campaign
```

```
SET current_amount = current_amount + 150.00
```

```
WHERE campaign_id = 1;
```

QUR_005: Update Donor Profile

Purpose: Allows a donor to change their phone number or other details.

Context: User profile management

```
UPDATE Donor
```

```
SET phone = '+201098765432'
```

```
WHERE donor_id = 1;
```

iii. Delete Data (Removing Records)

QUR_006: Cancel Recurring Donation

Purpose: Stops a scheduled recurring donation.

Context: Instead of deleting the record entirely (which destroys history), we often update status, but if a "hard delete" is required for the project, this removes the schedule

```
DELETE FROM Recurring_Donation  
WHERE recurring_id = 5 AND donor_id = 1;
```

QUR_007: Remove an Empty Campaign

Purpose: Deletes a campaign that was created by mistake but has received zero donations.

Context: Campaign management cleanup.

```
DELETE FROM Campaign  
WHERE campaign_id = 10 AND current_amount = 0.00;
```

iv. Select Queries (Complex Retrieval)

QUR_008: Donor Donation History (JOIN)

Purpose: Retrieves a detailed list of all donations made by a specific donor, including the campaign name and charity name.

Context: Used for the "My Donations" page.

```
SELECT  
    d.donation_id,  
    d.donation_date,  
    d.amount,  
    c.title AS campaign_name,  
    ch.name AS charity_name,  
    d.transaction_status  
FROM Donation d  
JOIN Campaign c ON d.campaign_id = c.campaign_id  
JOIN Charity ch ON c.charity_id = ch.charity_id  
WHERE d.donor_id = 1  
ORDER BY d.donation_date DESC;
```

QUR_009: Charity Dashboard Analytics (AGGREGATE & GROUP BY)

Purpose: Shows a charity the total amount raised and number of donations for each of their campaigns.

Context: Used for the Charity Dashboard to track performance

```
SELECT
  c.title,
  COUNT(d.donation_id) AS total_donations_count,
  SUM(d.amount) AS total_amount_raised,
  (c.current_amount / c.target_amount) * 100 AS progress_percentage
FROM Campaign c
LEFT JOIN Donation d ON c.campaign_id = d.campaign_id
WHERE c.charity_id = 1
GROUP BY c.campaign_id, c.title, c.current_amount, c.target_amount;
```

QUR_010: Filter Active Campaigns (SUBQUERY)

Purpose: Finds all active campaigns in the 'Health' category that have raised more than the average amount of all campaigns.

Context: Advanced filtering or "Featured Campaigns" logic

```
SELECT title, description, current_amount
FROM Campaign
WHERE category = 'Health'
AND status = 'Active'
AND current_amount > (SELECT AVG(current_amount) FROM Campaign);
```

Requirement ID	Requirement Description	Corresponding Query ID(s)
REQ_001	Donor Registration and Profile Creation	QUR_001, QUR_005
REQ_002	Charity Registration and Profile Creation	QUR_002 (adapted for Charity insert)
REQ_003	Charity and Campaign Browsing/Search	QUR_010
REQ_004	Donation Processing	QUR_003, QUR_004
REQ_006	Campaign Creation and Management	QUR_002, QUR_007
REQ_008	Recurring Donations Setup	QUR_006
REQ_009	Charity Dashboard for Statistics	QUR_009
REQ_010	Donor Donation History	QUR_008